

HW-2

Packages

Question 1

```
install.packages("aRtsy")
```

Install the `aRtsy` package

Question 2

```
library(aRtsy)
```

The Collatz conjecture starts with any positive integer n , if n is even, divide it by two; otherwise, multiply it by 3 and add 1. Repeats this process with the new n value. Eventually, n will reach to 1.

Let $f(n)$ be the next number obtained from n :
$$f(n) = \begin{cases} n/2, & \text{if } n/2 = 0, \\ 3n + 1, & \text{if } n/2 = 1. \end{cases}$$